WEB DESIGN FOR EVERYBODY · CAPSTONE

Table Tennis Central

Project report by Le Bao Hung

1. Overview

Table Tennis Central is a three page informational site about table tennis. It explains why the sport is worth playing, what equipment a beginner needs, and how a match is scored. Every illustration on the site is an original SVG drawn for this project.

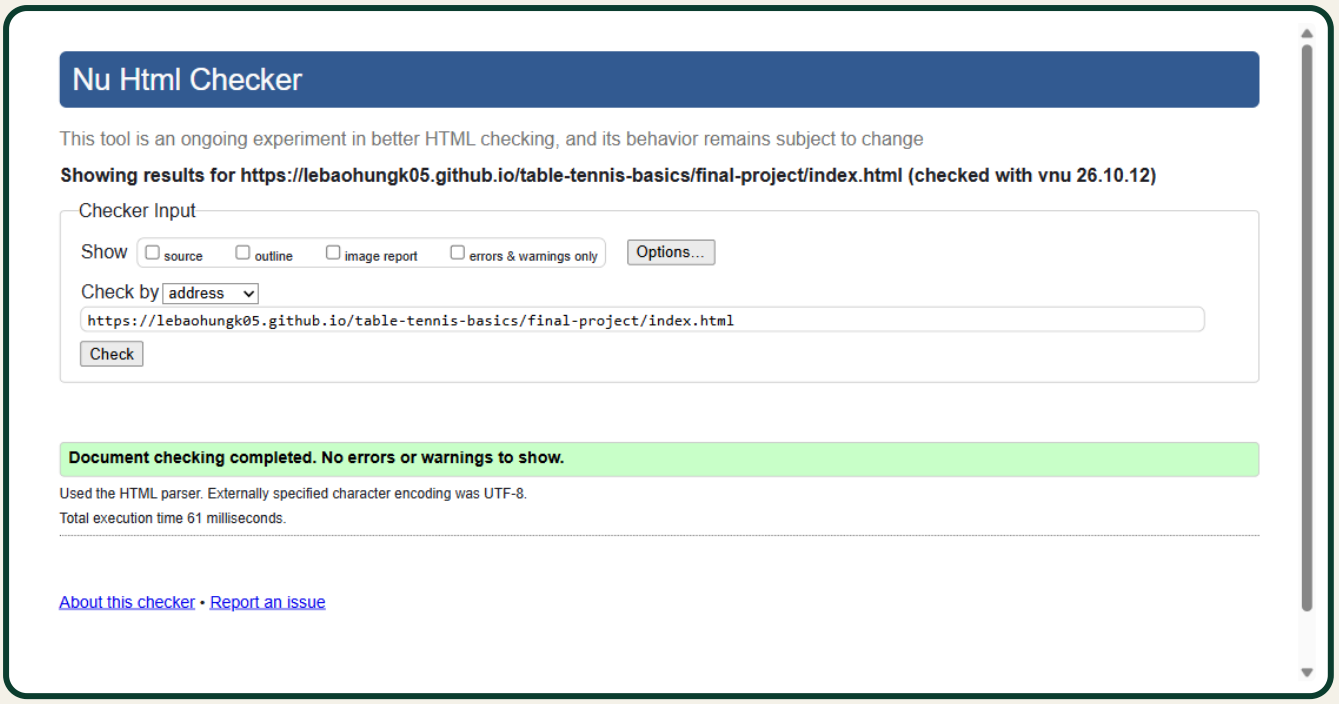
> **Live site:** https://lebaohungk05.github.io/table-tennis-basics/final-project/
>
> **Source code:** github.com/lebaohungk05/table-tennis-basics/final-project

Pages on the site

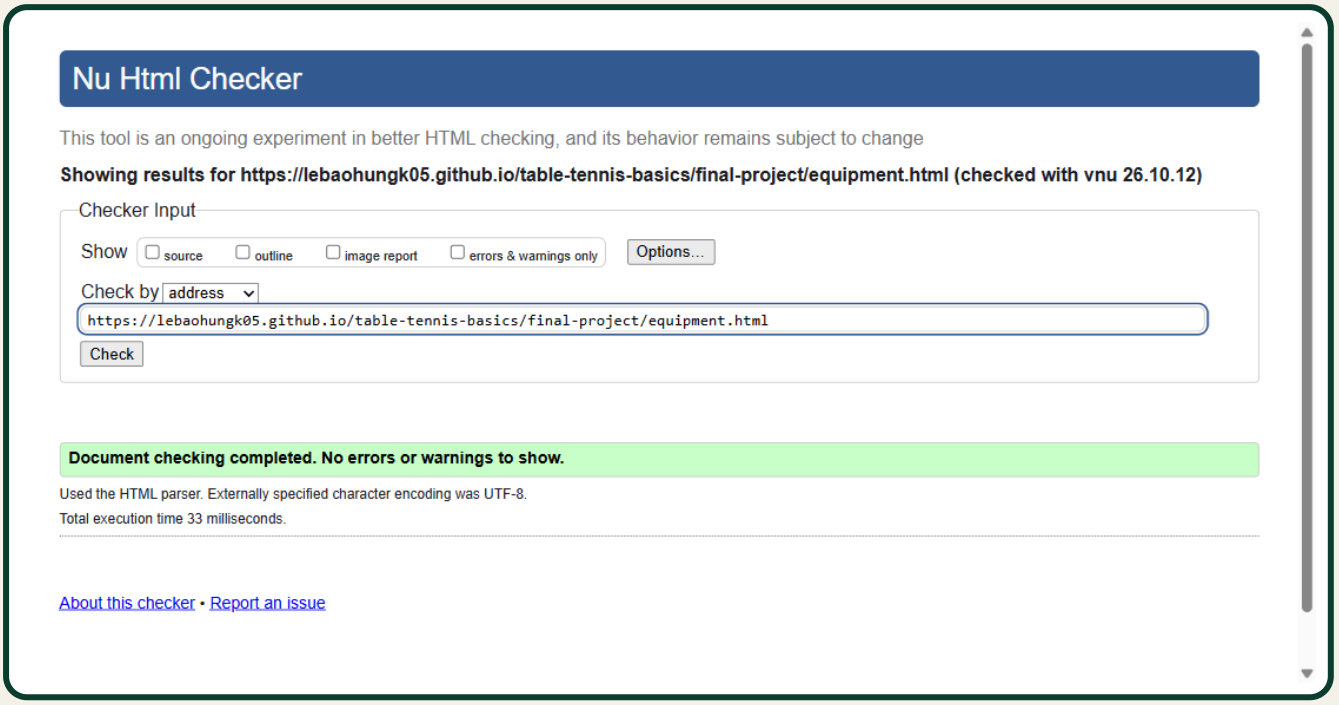
Page	Content	Images
Overview	Hero banner, key facts, five-image photo gallery, three “Why Play?” cards	7
Equipment	Five-item kit gallery, three paddle-type cards, ITTF specification table	9
Rules	Serve, scoring and expedite sections with diagrams, rally-call table, common faults	4

2. W3C Validator Compliance

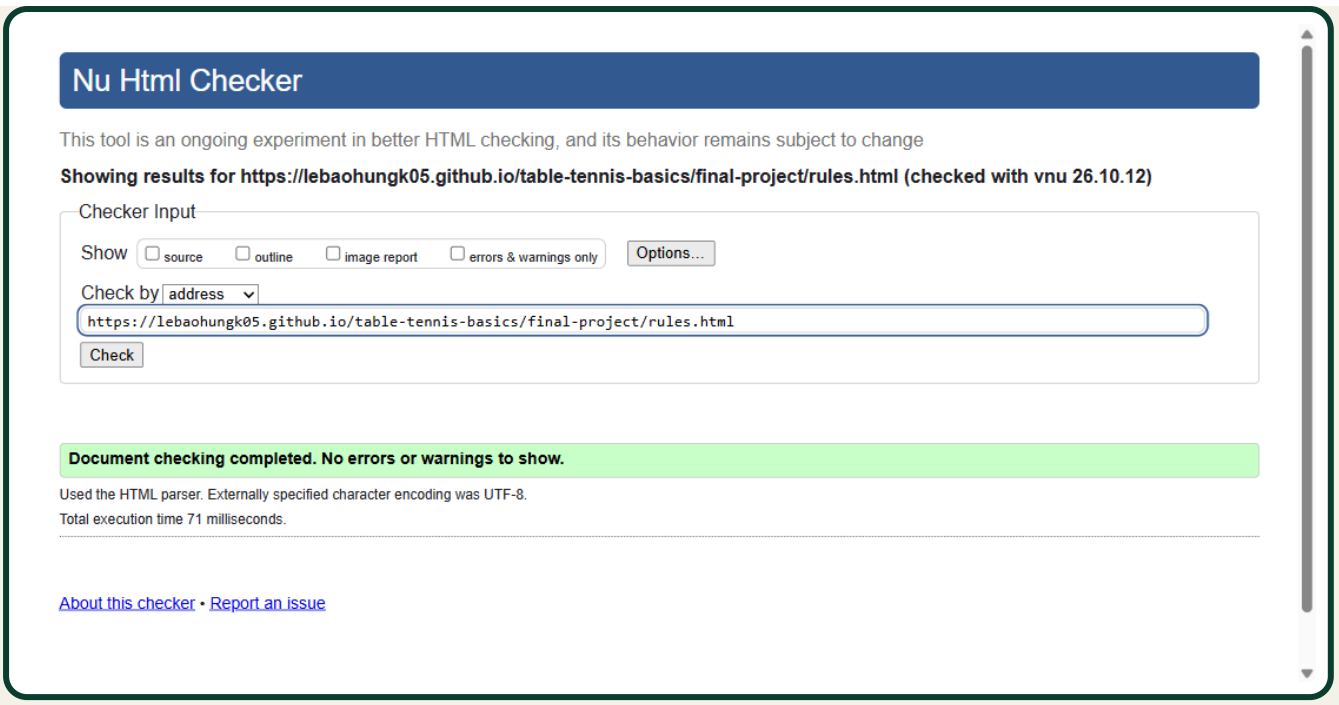
Each page was checked by address on the W3C Nu Html Checker, and the stylesheet on the W3C CSS Validation Service. All four report zero errors and zero warnings. The screenshots below are the validators' own result pages for the live URLs.



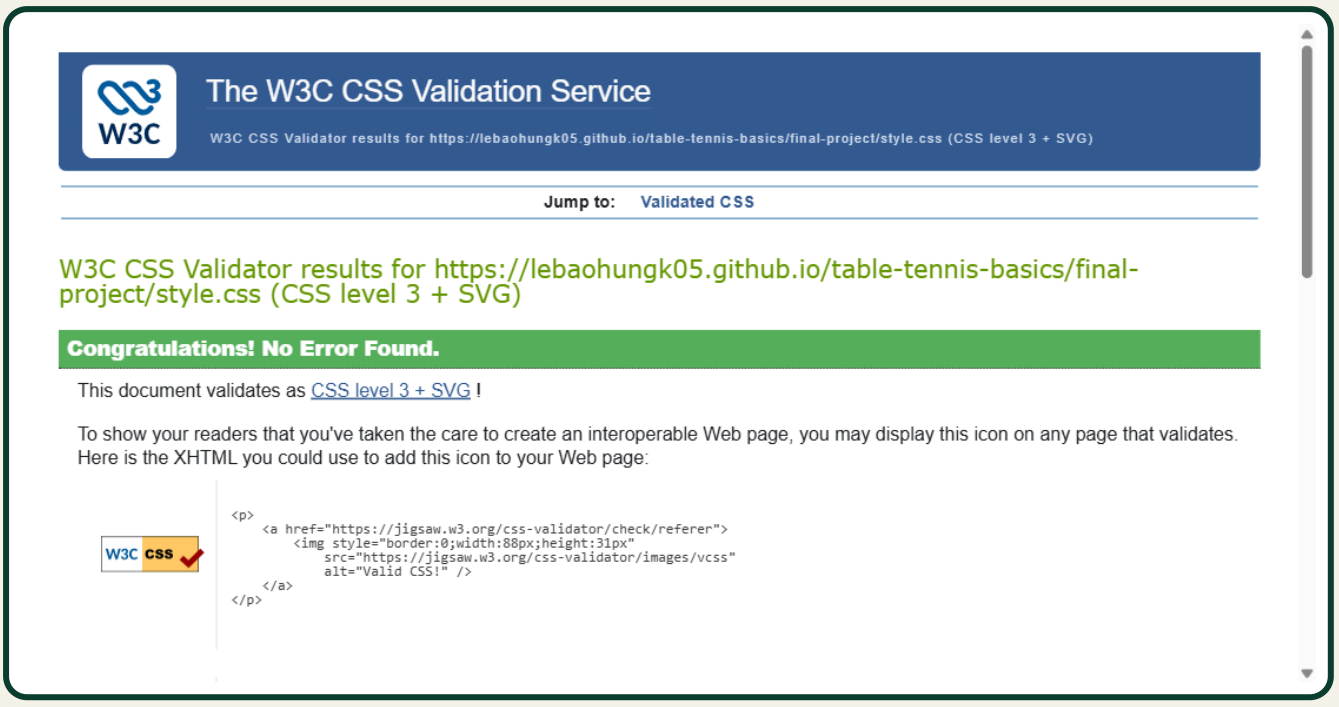
index.html — no errors or warnings



equipment.html — no errors or warnings



rules.html — no errors or warnings

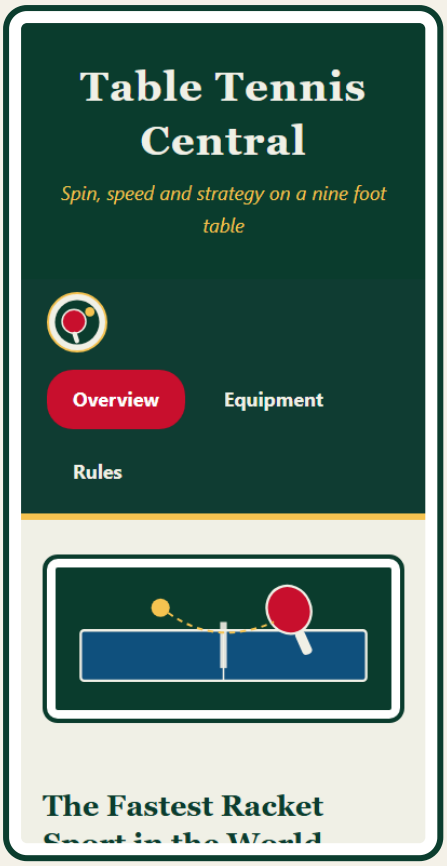


style.css — “Congratulations! No Error Found.”

As an extra accessibility check, all three pages were also run through the axe-core 4.10.2 engine (the engine behind the aXe browser extension), which reported **0 violations** on every page.

3. Responsive Design

The layout is mobile-first. On narrow screens the text sections stack in one column and the gallery shows two cards per row. A single media query at 720px widens the gallery to three columns and puts the text and the fact list side by side. The navigation uses `flex-wrap`, so its links reflow by themselves on small screens. None of the three widths below produce horizontal scrolling.



Mobile · 375px



Tablet · 768px



Desktop · 1280px

Photo gallery at each width

Photo Gallery

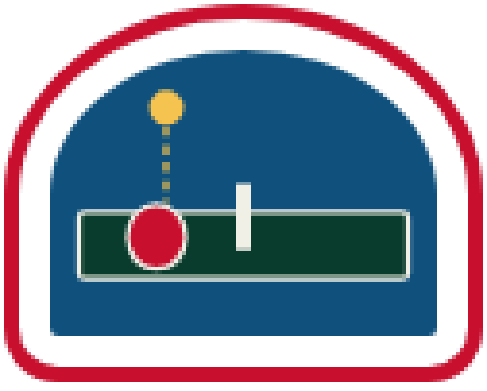
Five moments that show what a rally actually looks like.



Forehand drive: the engine of attacking play



Backhand block: absorbing an opponent's loop



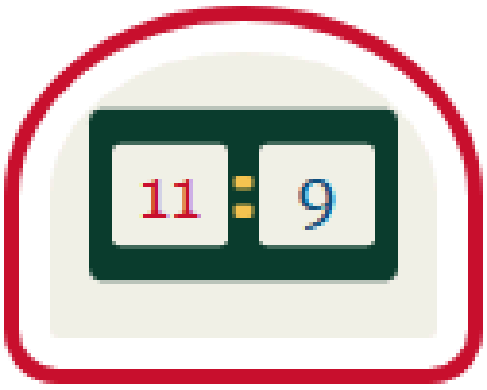
The legal serve: a



Over the net: the

16 cm vertical toss

ball must clear 15.25 cm



11–9: a game must be won by two clear points

375px: 2 columns

Photo Gallery

Five moments that show what a rally actually looks like.



Forehand drive: the engine of attacking play



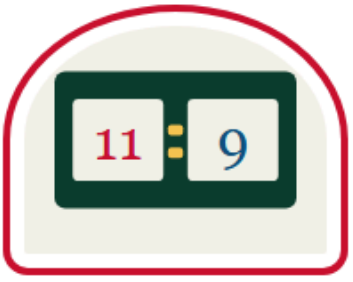
Backhand block: absorbing an opponent's loop



The legal serve: a 16 cm vertical toss



Over the net: the ball must clear 15.25 cm



11–9: a game must be won by two clear points

768px: 3 columns

Photo Gallery

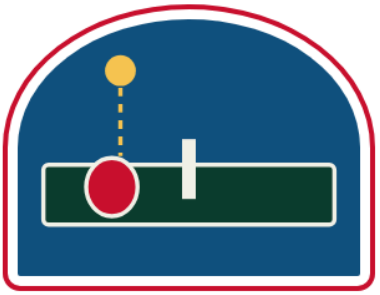
Five moments that show what a rally actually looks like.



Forehand drive: the engine of attacking play



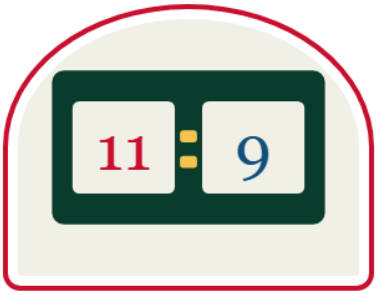
Backhand block: absorbing an opponent's loop



The legal serve: a 16 cm vertical toss



Over the net: the ball must clear 15.25 cm



11–9: a game must be won by two clear points

1280px: 3 columns

```
img {
  max-width: 100%;
  height: auto;
}

.gallery {
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 24px;
}

@media screen and (min-width: 720px) {
  .gallery {
    grid-template-columns: repeat(3, 1fr);
  }

  .split {
    grid-template-columns: 2fr 1fr;
  }
}
```

4. Styling

All three pages share one external stylesheet, linked from each page with `<link rel="stylesheet" href="style.css">`. The design uses a fixed five-colour palette taken from a table tennis venue: court green, table blue, paddle red, ball yellow and a warm off-white background.

`#0b3d2e` court green `#12527f` table blue `#c8102e` paddle red

`#f6c453` ball yellow `#f4f1e8` background

Colour contrast of the text pairs used (WCAG AA needs 4.5:1 for body text)

Text / background	Where	Ratio
#f4f1e8 on #0b3d2e	Header, nav, footer	10.8 : 1

Text / background	Where	Ratio
#0b3d2e on #f6c453	Nav hover, skip link	7.6 : 1
#12527f on #f4f1e8	Body links	7.2 : 1
#c8102e on #f4f1e8	Link hover	5.2 : 1

Typography and the box model

Headings use Georgia for a classic sports-programme feel, and body text uses Segoe UI at 18px with a 1.7 line height so paragraphs stay easy to read. Every content image is framed with the box model: padding gives it a white mat, a border outlines it, and border-radius rounds the corners.

```
.hero img,
figure img,
.card img,
.split > img {
  display: block;
  width: 100%;
  padding: 8px;
  background-color: #ffffff;
  border: 4px solid #0b3d2e;
  border-radius: 14px;
}
```

5. Four Extras

Beyond the base requirements, the site adds these four features.

1 Skip to Main Content link

The first element in every page is a link that jumps keyboard and screen-reader users straight past the header and navigation. It is moved off screen with `position: absolute` and slides into view only when it receives keyboard focus. The screenshot was taken right after pressing `Tab` once.



Visible only on focus

```
.skip-link {
  position: absolute;
  top: -100px;
  left: 16px;
  transition: top 0.25s ease-in-out;
}

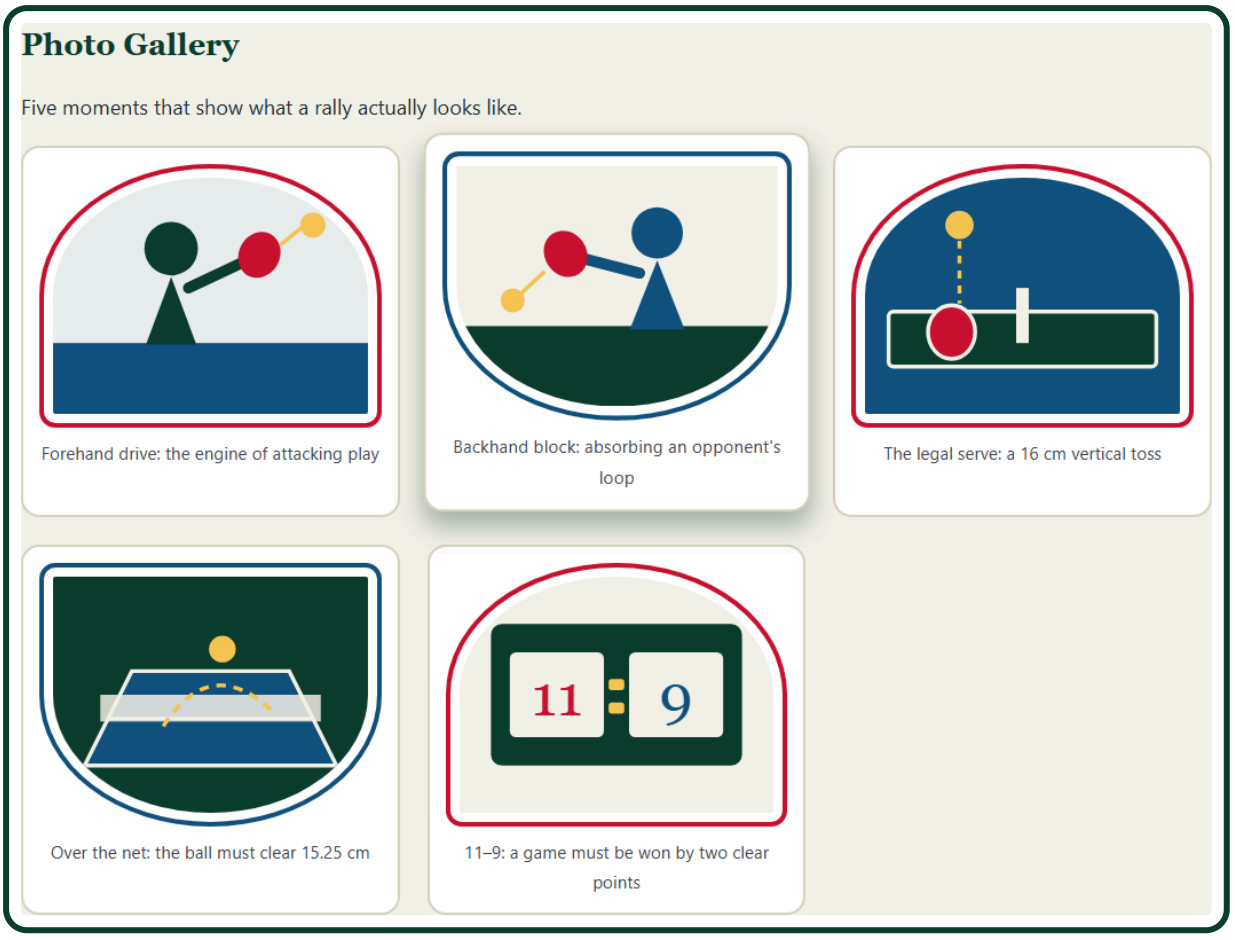
.skip-link:focus {
  top: 0;
  outline: 3px solid #c8102e;
}
```

2 Hover and focus animations with CSS transitions

Interactive elements respond with short transitions instead of instant jumps. Navigation links lift and change to yellow, gallery cards rise and gain a shadow, and the “Why Play?” cards tilt slightly. The navigation effect is attached to `:focus` as well as `:hover`, so keyboard users see it too.



Navigation link on hover



Gallery card on hover

```
.main-nav a {
  transition: background-color 0.3s ease, color 0.3s ease, transform 0.3s ease;
}

.main-nav a:hover,
.main-nav a:focus {
  background-color: #f6c453;
  color: #0b3d2e;
  transform: translateY(-4px);
}

.gallery figure:hover {
  transform: translateY(-8px) scale(1.02);
  box-shadow: 0 12px 22px rgba(11, 61, 46, 0.35);
}
```

3 CSS Grid and Flexbox layout

Grid handles the two-dimensional layouts: the photo galleries and the text-plus-facts split. Flexbox handles the one-dimensional rows: the navigation bar and the card rows. Both use `gap` for spacing, so the layout needs no float hacks or margin tricks.

```
.main-nav {
  display: flex;
  flex-wrap: wrap;
  align-items: center;
  justify-content: space-between;
  gap: 16px;
}

.card-row {
  display: flex;
  flex-wrap: wrap;
  gap: 24px;
}

.card {
  flex: 1 1 280px;
}
```

4 Advanced `nth-child` selectors

Instead of adding a class to every element, `nth-child` patterns give repeated items alternating styles. Gallery images alternate between arch and bowl shapes in red and blue. Cards rotate through three accent colours with `3n + 1`, `3n + 2` and `3n + 3`. Fact lists and table rows get zebra striping.



Card accents set by `:nth-child(3n + k)`

```
.gallery figure:nth-child(odd) img {
  border-radius: 50% 50% 14px 14px;
  border-color: #c8102e;
}

.gallery figure:nth-child(even) img {
  border-radius: 14px 14px 50% 50%;
  border-color: #12527f;
}

.card:nth-child(3n + 1) { border-top: 10px solid #c8102e; }
.card:nth-child(3n + 2) { border-top: 10px solid #12527f; }
.card:nth-child(3n + 3) { border-top: 10px solid #f6c453; }

.score-table tbody tr:nth-child(even) {
  background-color: #efeadc;
}
```